Monthly Sales Trend

Completed task brief • Sample Superstore dataset • Excel + Python workflow

1. Description & Objective

Sales transactions were grouped by month to summarize revenue and visualize how sales changed over time. The task demonstrates date conversion, chronological sorting, monthly aggregation, and a line chart.

2. Dataset & Tools

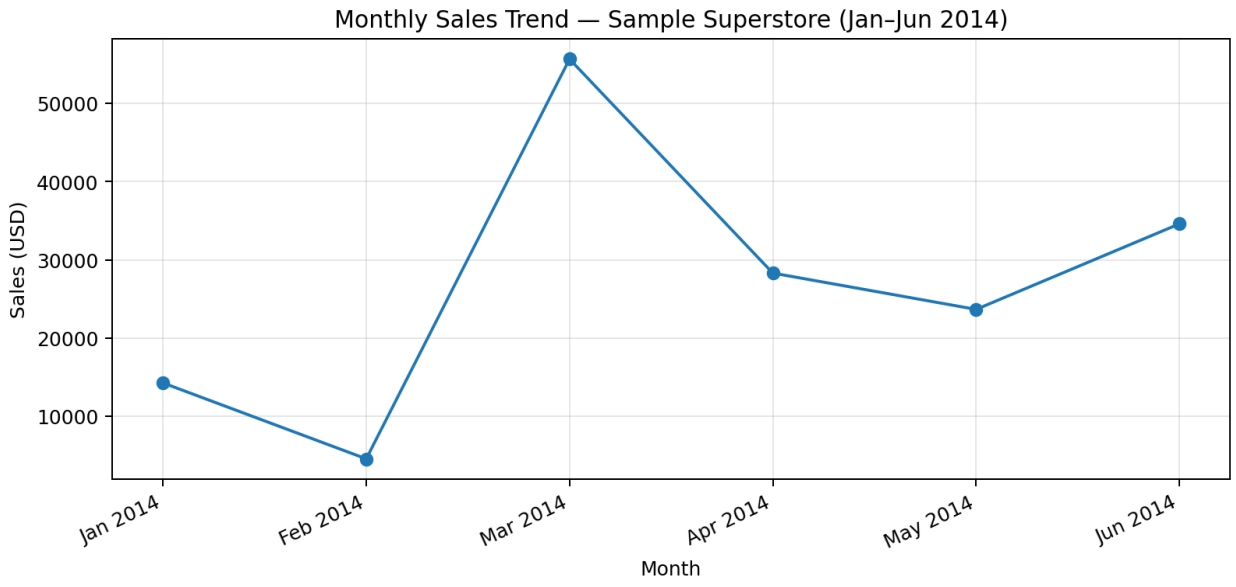
Dataset: Sample Superstore sales data (fictional educational dataset). The public analysis cited below reports 9,994 rows and provides the first six monthly aggregate values for 2014. This PDF uses those exact published values; it does not invent the remaining monthly totals.

Tools: Excel (PivotTable / line chart) and Python (pandas, Matplotlib).

3. Monthly Sales Summary

Month	Total Sales (USD)
Jan 2014	$14,236.90
Feb 2014	$4,519.89
Mar 2014	$55,691.01
Apr 2014	$28,295.35
May 2014	$23,648.29
Jun 2014	$34,595.13

4. Line Chart



5. Key Findings

- March 2014 had the highest sales in this six-month extract: $55,691.01.
- February 2014 had the lowest sales in this extract: $4,519.89.
- Sales rose sharply from February to March, then declined in April and May before recovering in June.
- This is only the first six months of the published monthly table. It supports a partial trend view, not a full 2014–2017 conclusion.

6. Method / Mini Guide

1) Parse Order Date as a date. 2) Create a Year-Month field. 3) Group by Year-Month and sum Sales. 4) Sort by date ascending. 5) Insert a line chart with Year-Month on the X-axis and Total Sales on the Y-axis.

Python example to reproduce the full analysis after downloading the complete CSV:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("superstore_sales.csv")
df["Order Date"] = pd.to_datetime(df["Order Date"], errors="coerce")
df = df.dropna(subset=["Order Date", "Sales"])
monthly = (df.assign(Month=df["Order Date"].dt.to_period("M").dt.to_timestamp())
             .groupby("Month", as_index=False)["Sales"].sum()
             .sort_values("Month"))
print(monthly.to_string(index=False))
monthly.plot(x="Month", y="Sales", kind="line", marker="o",
             title="Monthly Sales Trend", legend=False)
plt.xlabel("Month"); plt.ylabel("Sales (USD)")
plt.tight_layout(); plt.show()
```

7. Data Source & Transparency

Source for the monthly values and aggregation method: Camila Pattaro, “Superstore Sales Analysis with R” (published 23 Feb 2025), which reports the Sample Superstore dataset and monthly totals for Jan–Jun 2014. Source page: https://rstudio-pubs-static.s3.amazonaws.com/1289787_dd816b89a2d4483d8b2b6811230c5611.html. Dataset overview/download page: https://www.mathewkanalytics.com/datasets/superstore-sales/. Note: The underlying dataset is a fictional sample, not actual company sales. Only the first six published monthly values are included in the summary table here; download the complete CSV and run the code above for all months.